

Divide the 2 eqns give answer D.

- 9 D** Frictional force on the object provides the centripetal force $mr\omega^2$. Both objects have the same angular velocity and same mass, but the centripetal force required for Q is larger due to larger radius. When the centripetal force required exceeds the frictional force available, Q starts to slide.

- 10 D** $\frac{1}{2}mv^2 = \frac{1}{2}m\omega^2r^2$